

Entities interact differently with nether portals

MC - 295395 Block states

Status	Resolved / Fixed
Affected	1.21.5 Pre-release 2
Fixed in	25w16a
Reported	2025-03-15

STEPS TO REPRODUCE

1. Launch Minecraft version 25w04a.
2. Open a world.
3. Build a nether portal.
4. Enter the nether portal and wait until arriving in the Nether.
5. Return through the nether portal to the original dimension.
6. Place a boat a few blocks away from the portal.
7. Get in the boat.
8. Press F3 + B to show hitboxes, then switch to the third-person view with F5.
9. Slowly ride the boat into the nether portal.
10. Observe that the boat is not teleported until its hitbox touches or intersects the portal blocks' hitbox.

OBSERVED BEHAVIOUR

In the affected versions, an entity is not teleported when its hitbox merely occupies the same block space as a Nether portal block. The entity's hitbox must physically intersect the portal block's portal area before the portal detects it and initiates teleportation.

As a result, entities enter farther into the portal before teleporting than they did in 1.21.4. This can be observed with boats, minecarts, horses, and dropped items. The changed detection distance also affects mechanisms that rely on entities passing through portals, such as basic minecart- or boat-based chunk loaders, which take longer to transfer the entity through the portal.

EXPECTED BEHAVIOUR

Entities such as boats, minecarts, horses, and dropped items should teleport through a Nether portal as soon as their hitboxes are within the same block space as the portal blocks. Their hitboxes should not need to physically intersect the portal block hitboxes, matching the behavior of previous versions and allowing portal-dependent mechanics and contraptions to function consistently.

ENVIRONMENT

Minecraft Java Edition Version 1.21.5 Pre-release 2

ORIGINAL DESCRIPTION

Affected versions: 25w04a, 25w05a, 25w06a, 25w07a, 25w08a, 25w09a, 25w09b, 25w10a, 1.21.5-pre1, 1.21.5-pre2

Summary: All entities hitbox needs to be intersecting the nether portal blocks hitbox in order to teleport to the other dimension rather than just being in the same block space as the portal block like before. In the latest release (1.21.4) an entity's hitbox needed to be within the block space of a nether portal block to be teleported, in the latest snapshot their hitbox needs to intersect the portal blocks hitbox, this change wasn't mentioned in the changelog and can break some redstone contraptions such as the basic minecart or boat based chunk loaders.

Steps to Reproduce:

- Build a nether portal

Load the nether side then come back. (This removes the initial lag from loading the nether for the first time)

- Place down a boat a few blocks away from the portal and get in it.

- Turn on hitboxes (F3 + B).
- Slowly ride the boat into the portal and keep an eye on where the edge of the hitbox is before being teleported (F5 helps).
- See that it doesn't get teleported until touching the portal blocks hitbox.

See attached videos below of before/after with minecart example causing the minecart to take longer to go through the portal

Observed Results: Entities need to be intersecting the nether portal block to be teleported.

Expected Results: Entities should only need to be within the same block space as the nether portal block to be teleported like in previous versions.

Further information 05/03/25: after further testing i have found that the issue happens from a change to either nether portal blocks or entities (as it seems to affect boats/horses/items as well) where their hitbox now needs to touch the hitbox of the nether portal block rather than just being within the same block space as the portal block

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-295395>